Calculus I -- Midterm Examination

Spring 2026 -- 90 minutes -- 100 points

Name: *(write name here)* **Student ID:** *(write ID here)*

Instructions. Answer all five questions in the space provided. Show your work; partial credit is awarded for correct method even when the final number is wrong. Calculators are *not* permitted. A blank Answer Key appears at the end of the booklet for grader use only – students should not write in that section.

Question 1. Limits (15 points)

Evaluate the limit of (x squared minus four) divided by (x minus two) as x approaches two. Show each algebraic step, especially the factoring that removes the indeterminate form.

Question 2. Differentiation (20 points)

Let f(x) equal x cubed times sine of x. Compute the derivative f-prime of x using the product rule, then evaluate f-prime at pi.

Question 3. Chain rule (20 points)

Differentiate g(x) equal to the square root of (one plus cosine of two-x) with respect to x. Simplify your answer until no double-angle identities remain.

Question 4. Definite integration (20 points)

Evaluate the definite integral, from zero to pi-over-two, of sine of x times cosine of x with respect to x. A u-substitution will make this routine – identify it explicitly before integrating.

Question 5. Application (25 points)

A spherical balloon is being inflated so that its radius increases at a constant rate of 2 cm/s. At the instant the radius reaches 10 cm, how fast is the volume increasing? Recall that the volume of a sphere is four-thirds pi r-cubed.

Answer Key

This page is for grader use only. Students should not write here.

Q1. Factor the numerator as (x - 2)(x + 2). The factor of (x - 2) cancels, leaving the limit of (x + 2) as x approaches 2, which is 4.

Q2. By the product rule, f-prime(x) equals 3 x-squared sine(x) plus x-cubed cosine(x). Evaluating at x = pi gives f-prime(pi) = 3 pi-squared times 0 plus pi-cubed times (-1) = -pi-cubed.

Q3. Let u = 1 + cos(2x), so g = sqrt(u). Then g-prime(x) = (1 / (2 sqrt(u))) times (-2 sin(2x)) = -sin(2x) / sqrt(1 + cos(2x)). Using 1 + cos(2x) = 2 cos-squared(x) and sin(2x) = 2 sin(x) cos(x), this simplifies to g-prime(x) = -sqrt(2) sin(x) times sgn(cos(x)).

Q4. With u = sin(x), du = cos(x) dx, the integral becomes the integral from 0 to 1 of u du, which is 1/2.

Q5. Differentiating V = (4/3) pi r-cubed gives dV/dt = 4 pi r-squared dr/dt. With r = 10 cm and dr/dt = 2 cm/s, dV/dt = 800 pi, approximately 2513.3 cubic centimetres per second.